

MLCA

Hyevin Shin

```
## -----  
##' [libraries]  
## -----
```

```
library(tidyverse)  
library(readxl)  
library(tidycensus)  
library(multilevLCA)  
library(psych)  
library(tigris)  
library(nnet)  
library(VGAM)  
library(gt)  
library(car)  
library(writexl)
```

```
rm(list = ls())
```

```
## -----  
##' [IPEDS/Carnegie Classification Data]  
## -----
```

```
# load IPEDS data and merge with the Carnegie Classification
```

```
ipeds24 <- read_csv("../data/sources/hd2024.csv")  
ipeds_cost <- read_csv("../data/sources/cost1_2024.csv")  
#ipeds_schoolsize <- read_csv("../data/sources/ef2024a.csv") #enrollment Fall 2024  
carnegie <- read_excel("../data/sources/2025-Public-Data-File.xlsx", sheet = 2)  
ic2024 <- read_csv("../data/sources/ic2024.csv")  
sfa <- read_csv("../data/sources/sfa2324.csv")  
adm <- read_csv("../data/sources/adm2024.csv")
```

```

enroll <- read_csv("../data/sources/effy2024.csv") %>%
  filter(EFFYALEV == 3) %>% #degree/certificate-seeking undergraduates
  select(UNITID, EFFYALEV, EFYTOTLT, EFYAIANT, EFYBKAAT, EFYNHPIT, EFYWHITT, EFYHISPT, EFYAS)
#ipeds_distance <- read_csv("../data/sources/effy2024_dist.csv") %>%
# filter(EFFYDLEV == 2) %>% #undergrad total
# select(UNITID, EFFYDLEV, EFYDETOT, EFYDEEXC)
ipeds_dist <- read_csv("../data/sources/ef2024a_dist.csv") |>
  filter(EFDELEV == 2) |> #undergrad total
  select(UNITID, EFDELEV, EFDETOT, EFDEEXC, XEFDEEXC)
ipeds_learner <- read_csv("../data/sources/ef2024b.csv") |>
  filter(LSTUDY == 2, EFBAGE %in% c(1, 7)) |> #undergrad total / all age categories total
#number of institutions (= UNITID): 2,487
  select(UNITID, EFBAGE, EFAGE06, EFAGE09) |>
  pivot_wider(names_from = EFBAGE,
              values_from = c(EFAGE06, EFAGE09))

#Let's replace this with the size variable in IPEDS (instsize)
# ipeds_schoolsize <- ipeds_schoolsize |>
#   filter(EFALEVEL == 1) |> #Level of student (1 == all students total)
#   select(UNITID, EFTOTLT) |> #Grand total
#   mutate(size = case_when(
#     EFTOTLT <= 3000 ~ 0, #small
#     EFTOTLT <= 10000 ~ 1, #medium
#     EFTOTLT > 10000 ~ 2, #large
#     TRUE ~ NA_real_))

carnegie <- carnegie |>
  select(unitid, hsi)

# merge files
data <- ipeds24 %>%
  left_join(carnegie, by = c("UNITID" = "unitid")) %>%
  left_join(ipeds_cost, by = "UNITID") |>
  #left_join(ipeds_schoolsize, by = "UNITID") %>%
  left_join(ic2024, by = "UNITID" ) %>%
  left_join(sfa, by = "UNITID") %>%
  left_join(adm, by = "UNITID") %>%
  left_join(enroll, by = "UNITID") |>
  left_join(ipeds_dist, by = "UNITID") |>
  left_join(ipeds_learner, by = "UNITID")

```

```
# select variables and create a dataset for analysis
x <- data |>
  rename_all(tolower) |> #converts all column names in a data frame to lowercase
  select(unitid, instnm, city, stabbr, fips, countycd, countynm, cbsa, cbsatype, csa, ober
    locale, zip, longitud, latitude,
    control, iclevel, instcat, deggrant, instsize,
    hloffer,
    carnegieic, carnegiesaec, carnegiealf, carnegieapm, carnegiersch,
    hbcu, hsi, tribal,
    room, relaffil, distnced, athassoc,
    admssn, applcn, openadmp, loan_a, tuition1, upgrntp,
    eftytotlt, efyaiant, efybkaat, efyhnpit, efywhitt, efyhispt, efyasiat,
    efdelev, efdetot, efdeexc, xefdeexc, slo7, efage06_1, efage06_7, efage09_1, efage

# assign to "data"
data <- x
```

```
x <- data %>%
  filter(fips <= 56) |> # Keep only 50 states and DC
  filter(cbsa != -2) |> # Remove colleges not located in CBSA
  # filter(!is.na(control), control != 3) |> # 3: private for-profit
  filter(!is.na(iclevel), iclevel != 3) |> # 3: less than 2 years
  filter(!is.na(carnegiealf), carnegiealf != 6) %>% # 6: graduate-focused
  filter(instcat >= 2 & instcat <= 4) # 1: degree-granting, graduate with m
  #filter(carnegieapm == 1) # 5-6: non-degree-granting
  # Keep only colleges classified as "I"

data <- x
```

```
## -----
##' [Data Cleaning]
## -----

x <- data |>
  mutate(
    public = case_when(
      control == 2 ~ 0, # 0: private
      control == 1 ~ 1, # 1: public
      control == 3 ~ 2, # 2: for-profit
      TRUE ~ NA_real_),
```

```

twoyear = case_when(
  iclevel == 1 ~ 0,          # 0: 4 or more
  iclevel == 2 ~ 1,          # 1: at least 2 & less than 4
  TRUE ~ NA_real_),

# msi = case_when(          # 0: non of them
#   hbcu == 1 ~ 1,          # 1: HBCU
#   hsi == 1 ~ 2,          # 2: HSI
#   tribal == 1 ~ 3,       # 3: TCU
#   hbcu == 2 & hsi == 0 & tribal == 2 ~ 0,
#   is.na(hbcu) | is.na(hsi) | is.na(tribal) ~ NA_real_,
#   TRUE ~ NA_real_),

msi = case_when(
  hbcu == 1 | hsi == 1 | tribal == 1 ~ 1,
  TRUE ~ 0),

rurality = case_when(
  locale %in% c(31,32,33,41,42,43) ~ 0,    # 0: Town + Rural
  locale %in% c(21,22,23) ~ 1,            # 1: Suburb
  locale %in% c(11,12,13) ~ 2,            # 2: City
  TRUE ~ NA_real_),

rurality = factor(rurality,
  levels = c(0, 1, 2),
  labels = c("Town/Rural", "Suburb", "City")),

size = case_when(
  instsize == 1 ~ 0,          # 0: < 1,000
  instsize == 2 ~ 1,          # 1: 1,000 - 4,999
  instsize == 3 ~ 2,          # 2: 5,000 - 9,999
  instsize == 4 ~ 3,          # 3: 10,000 - 19,999
  instsize == 5 ~ 4,          # 4: >=20,000
  TRUE ~ NA_real_),

# tuition = case_when(
#   tuition1 == 0 & obereg != 0 ~ NA_real_,
#   tuition1 == 0 & obereg == 0 ~ 0,
#   tuition1 < 10000 ~ 0,      # 0: < $10,000 - 1334
#   tuition1 >= 10000 & tuition1 < 20000 ~ 1, # 1: $10,000 - $19,999 - 689
#   # tuition1 >= 20000 & tuition1 < 30000 ~ 2, # 2: $20,000 - $29,999
#   # tuition1 >= 20000 & tuition1 < 40000 ~ 2, # 3: $30,000 - $39,999 - 531

```

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# tuition1 >= 40000 & tuition1 < 50000 ~ 4,      # 4: $40,000 - $49,999
# tuition1 >= 40000 ~ 3,      # 5: > $50,000      - 384
# TRUE ~ NA_real_),      #      - 316

# tuitionnn = ntile(tuition1, 4) - 1,      # 0~3: quartiles

#check missing values: sum(is.na(ipeds_cost$TUITION1)) [1] 2,368 (= about 40% of the sample)

housing = case_when(
  room == 2 ~ 0,      # 0: no housing
  room == 1 ~ 1,      # 1: offer housing
  TRUE ~ NA_real_),

religious = ifelse(relaffil == -2, 0, 1),      # 0: no religious
      # 1: any religious

#We need to remove colleges that offer completely distance education from our sample
distance = case_when(
  distnced == 2 ~ 0,      # 0: not completely distanced learning
  distnced == 1 ~ 1,      # 1: completely distanced learning
  TRUE ~ NA_real_),

online_perc = case_when(
  is.na(efdetot) ~ NA_real_,
  is.na(efdeexc) & xefdeexc == "A" ~ 0,      # "A": not applicable
  is.na(efdeexc) ~ NA_real_,      # other missing cases
  TRUE ~ efdeexc / efdetot),

# online = case_when(
#   is.na(online_perc) ~ NA_real_,
#   online_perc == 0 ~ 0,      # none      - 8
#   online_perc < 0.25 ~ 1,      # more than 0%, less than 25% - 1
#   online_perc < 0.50 ~ 2,      # 25% to less than 50%      - 7
#   online_perc <= 1 ~ 3,      # 50% to 100%      - 2
#   TRUE ~ NA_real_),      # NA      - 1

online = ntile(online_perc, 4) - 1,      # 0~3: quartiles

#check missing values: sum(is.na(ipeds_dist$EFDETOT)) [1] 0

evening = case_when(
  slo7 == 0 ~ 0,      # 0: no weekend/evening courses

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    slo7 == 1 ~ 1,                                     # 1: allows classes only on weekend
    TRUE ~ NA_real_),

#check missing values: sum(is.na(ic2024$SL07)) [1] 0

#      adult_perc = case_when(
#          is.na(efage09_7) ~ NA_real_,
#          is.na(efage09_1) ~ NA_real_,
#          TRUE ~ efage09_7 / efage09_1),

#      adult_perc = case_when(
#          is.na(efage09_7) ~ NA_real_,
#          is.na(efage09_1) ~ NA_real_,
#          TRUE ~ efage09_7 / efage09_1),

#      adult = ntile(adult_perc, 4) - 1,                # 0~3: quartiles

#check missing values: sum(is.na(ipeds_learner$EFAGE09_1)) [1] 0
#check missing values: sum(is.na(ipeds_learner$EFAGE09_7)) [1] 9

#      parttime_perc = case_when(
#          is.na(efage06_1) ~ NA_real_,
#          is.na(efage09_1) ~ NA_real_,
#          TRUE ~ efage06_1 / efage09_1),

#      parttime = case_when(
#          is.na(parttime_perc) ~ NA_real_,
#          parttime_perc == 0 ~ 0,                        # none - 73
#          parttime_perc < 0.25 ~ 1,                    # more than 0%, less than 25% - 83
#          parttime_perc < 0.50 ~ 2,                    # 25% to less than 50% - 34
#          parttime_perc <= 1 ~ 3,                      # 50% to 100% - 70
#          TRUE ~ NA_real_),                            # NA - 14

#      parttime = ntile(parttime_perc, 4) - 1,          # 0~3: quartiles
#check missing values: sum(is.na(ipeds_learner$EFAGE06_1)) [1] 0

sports = case_when(
    athassoc == 2 ~ 0,                                     # 0: not a member
    athassoc == 1 ~ 1,                                     # 1: member of National Athletic A
    TRUE ~ NA_real_),

loan = case_when(

```

```

is.na(loan_a) ~ NA_real_,
loan_a <= quantile(loan_a, 0.25, na.rm = TRUE) ~ 0,      # 0: lowest
loan_a <= quantile(loan_a, 0.50, na.rm = TRUE) ~ 1,      # 1: low
loan_a <= quantile(loan_a, 0.75, na.rm = TRUE) ~ 2,      # 2: high
TRUE ~ 3),                                               # 3: highest

```

Q. Since MLCA requires variables starting at 0, is it appropriate to divide loan amount into

```

asc = case_when(
  carnegiesaec == 2 ~ 0,      # 0: Higher Access, Lower Earnings
  carnegiesaec == 4 ~ 1,      # 1: Higher Access, Medium Earnings
  carnegiesaec == 6 ~ 2,      # 2: Higher Access, Higher Earnings
  carnegiesaec == 1 ~ 3,      # 3: Lower Access, Lower Earnings
  carnegiesaec == 3 ~ 4,      # 4: Lower Access, Medium Earnings
  carnegiesaec == 5 ~ 5,      # 5: Lower Access, Higher Earnings
  TRUE ~ NA_real_),

```

```

awardlev = case_when(
  carnegiealf %in% c(1,2) ~ 0,      # 0: associates
  carnegiealf == 3 ~ 1,             # 1: baccalaureate
  carnegiealf == 4 ~ 2,             # 2: undergrad / grad-Master's
  carnegiealf == 5 ~ 3,             # 3: undergrad / grad-Doctorate

```

```

is.na(carnegiealf) & (hloffer == 3 | hloffer == 4) ~ 0,
carnegiealf == -2 & (hloffer == 3 | hloffer == 4) ~ 0,
hloffer == 5 & instcat == 2 ~ 1,

```

```

TRUE ~ NA_real_),

```

```

#Check: x %>% filter(is.na(awardlev)) %>% View()

```

```

focus = case_when(
  carnegieapm == -2 ~ 0, # 0: Not classified
  carnegieapm == 1 ~ 1, # 1: Mixed
  carnegieapm == 2 ~ 2, # 2: Professions-focused
  carnegieapm == 3 ~ 3, # 3: Applied and career studies
  carnegieapm %in% c(4,5) ~ 4, # 4: Arts and sciences
  carnegieapm == 6 ~ 5, # 5: Business
  carnegieapm %in% c(8,9,10) ~ 6, # 6: Medical sciences/health
  carnegieapm == 11 ~ 7, # 7: STEM
  carnegieapm == 12 ~ 8, # 8: Theology
  TRUE ~ NA_real_),

```

```

research = case_when(
  carnegiersch == -2 ~ 0,           #0: No research designation
  carnegiersch == 3 ~ 1,           #1: Research Colleges and Universities
  carnegiersch == 2 ~ 2,           #2: High Spending
  carnegiersch == 1 ~ 3,           #3: Very High Spending
  TRUE ~ NA_real_),

cbsa = as.character(cbsa),

cbsatype = case_when(
  cbsatype == 2 ~ 0,   # Micropolitan
  cbsatype == 1 ~ 1,   # Metropolitan
  TRUE ~ NA_real_),

pell = case_when(
  upgrntp < 10 ~ 0,
  upgrntp >= 10 & upgrntp < 30 ~ 1,
  #upgrntp >= 20 & upgrntp < 30 ~ 2,
  upgrntp >= 30 & upgrntp < 50 ~ 2,
  # upgrntp >= 40 & upgrntp < 50 ~ 4,
  upgrntp >= 50 ~ 3,
  TRUE ~ NA_real_)

)

data <- x |>
  filter(distance == 0) %>%           # remove observations with only distance e
  group_by(cbsa) %>%
  mutate(inst_num = n_distinct(unitid),
         inst_num = as.numeric(inst_num)) %>%
  ungroup() %>%
  mutate(admrate = admssn/applcn) %>%
  # mutate(admrate = if_else(openadmp == 1, 1, admrate)) |>
  mutate(selectivity = case_when(
    openadmp == 1 ~ 0,
    admrate == 1 ~ 0,               #0: open
    admrate >= 0.75 & admrate < 1 ~ 1,   #1: less selective
    admrate >= 0.40 & admrate < 0.75 ~ 2, #2: selective
    admrate < 0.40 ~ 3,               #3: more selective
    TRUE ~ NA_real_),

```

```

    i_perc_black = efybkaat/efytotlt, #Percent of degree-seeking undergrad
    i_perc_white = efywhitt/efytotlt,
    i_perc_hisp = efyhispt/efytotlt,
    i_perc_nhpi = (efyaiant + efynhpit)/efytotlt,
    i_perc_asian = efyasiat/efytotlt,
    i_perc_nonwhite = 1 - i_perc_white,
    nonwhite = ntile(i_perc_nonwhite, 4),
    nonwhite = nonwhite - 1
  ) %>%
rename(region = obereg) %>%
mutate(cbsa = as.character(cbsa))

```

```

z <- read_csv("../data/sources/effy2024.csv") %>%
  filter(EFFYLEV == 2 | EFFYLEV == 4) %>%
  select(UNITID, EFFYALEV, EFYTOTLT) %>%
  mutate(EFFYALEV = if_else(EFFYALEV == 2, "undergrad", "grad")) %>%
  pivot_wider(id_cols = "UNITID",
              names_from = EFFYALEV,
              values_from = EFYTOTLT) %>%
  mutate(total_pop = undergrad + grad,
         per_undergrad = undergrad/total_pop,
         per_grad = grad/total_pop)

```

```

## -----
## [Census/ACS Data]
## -----

land_area <- core_based_statistical_areas(
  cb = TRUE,
  year = 2024) %>%
  select(GEOID, ALAND)

census <- get_acs(geography = "cbsa",
                 year = 2024,
                 variables = c("B01003_001E", # Total population
                              "B19013_001E", # Median Household Income in the Past 12 Months
                              "DP03_0119PE", # % families below poverty line
                              "DP03_0005PE", # % unemployed over 16 years old
                              "DP02_0068PE", # % bach or higher
                              "B03002_003E", # total white (alone, not hispanic)

```

```

        "B03002_004E", # total black (alone, not hispanic)
        "B03002_012E", # total hispanic
        "B03002_006E", # total asian (alone, not hispanic)
        "B03002_001E" # total by ethnicity
    ),
    output = "wide") %>%
left_join(land_area)

census <- census |>
  mutate(perc_white = B03002_003E/B03002_001E*100,
         perc_black = B03002_004E/B03002_001E*100,
         perc_hisp = B03002_012E/B03002_001E*100,
         perc_asian = B03002_006E/B03002_001E*100,
         tot_pop = B01003_001E/100000,
         pop_density = B01003_001E/(ALAND/2589988.10)) |> #population per sq mile
  rename(perc_ba = DP02_0068PE,
         med_income = B19013_001E,
         perc_pov = DP03_0119PE,
         perc_unemployed = DP03_0005PE) |>
  select(tot_pop, pop_density, perc_ba,
         med_income, perc_pov, perc_unemployed,
         perc_white, perc_black, perc_hisp, perc_asian,
         GEOID) |>
  mutate(GEOID = as.character(GEOID))

data <- data |>
  left_join(census, by = c("cbsa" = "GEOID"))

```

```

data_mlca <- data %>%
  select(unitid, instnm, stabbr, cbsa, cbsatype, region,
         public, twoyear, size, rurality,
         awardlev,
         religious, focus, housing, research, msi,
         inst_num, selectivity, pell, online, online_perc, evening, #adult, tuitionnn, partt,
         tot_pop, pop_density, perc_ba, med_income, perc_pov, perc_unemployed,
         perc_white, perc_black, perc_hisp, perc_asian,
         i_perc_white, i_perc_black, i_perc_hisp, i_perc_asian, i_perc_nonwhite, nonwhite) %>%
  as.data.frame()

#metropolitan, tuition,msi, distance,sports,research,

```

```

#describe(data_mlca)

x <- data_mlca %>%
  drop_na() %>%
  group_by(cbsa) %>%
  mutate(
    cbsa_id = cur_group_id()
  ) %>%
  as.data.frame()

# data_mlca %>% anti_join(x, by = "unitid") %>% View()

data_mlca <- x

write_rds(data_mlca, "../data/analytic_v2")

```

```

# Variables:
#   Y-          Columns with indicator variables for level 1 class membership (predictors of i
#               # public, two_year, size, msi, award level, research, selectivity

#   iT-         Number of lower-level latent classes

#   id_high-    Name of column with higher-level id
#               # CBSA

#   iM-         Number of higher-level latent classes

#   Z-          Columns with lower-level covariates (predictors of level 1 class membership)
#               # rurality, housing, religious, sports

#   Zh-         Columns with higher-level covariates (predictors of level 2 class membership)
#               # Variables- region, number of IHEs, total population, percent bachelors deg

# Level-1 indicators: Observed institution characteristics used to define the latent institut
# Level-1 covariates: Institution-level predictors that explain class membership
# Level-2 covariates: CBSA-level regional variables that explain differences in institution c

# Lower-level indicators based on framework for insitutional diversity in:
# Harris, M.S. An Empirical Typology of the Institutional Diversity of U.S. Colleges and Uni

```

```

#indicators
Y <- c(
  #systemic- structural and governance characteristics
  "public", "size", #"twoyear",

  #programmatic- degree level and academic disciplinary offerings
  "awardlev", "focus", "research",

  #procedural- how programs are offered and implemented (Harris included % of students enrol.
  #the institution / management? operation?
  "online", #ADD VARIABLE FOR THE PERCENTAGE OF STUDENTS WHO ARE EXCLUSIVELY ENROLLED IN ONLINE
  #categorized share of undergraduates enrolled exclusively in distance education
  "evening",
  # "tuitionnn",
  #(1)including tuition as quartiles: iT= 8 & iM= 2      (#obs = 1779)
  #(2)including tuition & parttime quartiles: iT= 8 & iM= 2    (#obs = 1779)

  #constituentia- "student demographics, background, preparation, and abilities" (Harris, 2000)
  "selectivity", "pell", "msi", "nonwhite" #, "adult", "parttime"
) #level-1 indicators

Z <- c("i_perc_white", "i_perc_black", "i_perc_hisp", "i_perc_asian", "housing", "rurality",

Zh <- c("cbsatype", "inst_num", "tot_pop", "pop_density", "perc_ba", "med_income", "perc_pov")

```

```

## -----
## Descriptive Statistics
## -----

# -----
# Level 1: Institution-level variables
# -----

# Categorical Level-1 variables
l1_cat <- c("public", "size", "awardlev", "focus", "research", "evening",
           "selectivity", "pell", "msi", "housing", "rurality", "religious")

# Continuous Level-1 variables
l1_cont <- c("online_perc", "i_perc_nonwhite", "i_perc_white",

```

```

      "i_perc_black", "i_perc_hisp", "i_perc_asian")

# Proportion for categorical variables
l1_categorical_desc <- data_mlca |>
  select(all_of(l1_cat)) |>
  mutate(across(everything(), as.character)) |>
  pivot_longer(cols = everything(),
               names_to = "Variable",
               values_to = "Category") |>
  count(Variable, Category, name = "N") |>
  group_by(Variable) |>
  mutate(Proportion = N / sum(N)) |>
  ungroup() |>
  select(Variable, Category, Proportion) |>
  mutate(Proportion = round(Proportion, 2))

l1_categorical_desc

```

```

# A tibble: 44 x 3
  Variable Category Proportion
  <chr>      <chr>      <dbl>
1 awardlev  0          0.47
2 awardlev  1          0.24
3 awardlev  2          0.13
4 awardlev  3          0.16
5 evening   0          0.43
6 evening   1          0.57
7 focus     0          0.02
8 focus     1          0.3
9 focus     2          0.33
10 focus    3          0.09
# i 34 more rows

```

```

# Mean, SD for continuous variables
l1_continuous_desc <- data_mlca |>
  summarise(across(all_of(l1_cont),
                    list(Mean = ~mean(., na.rm = TRUE),
                         SD = ~sd(., na.rm = TRUE)))) |>
  pivot_longer(everything(),
               names_to = c("Variable", ".value"),

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      names_pattern = "(.+)_ (Mean|SD)$") |>
mutate(across(c(Mean, SD), ~round(., 2)))

l1_continuous_desc

```

```

# A tibble: 6 x 3
  Variable      Mean    SD
  <chr>        <dbl> <dbl>
1 online_perc    0.17  0.22
2 i_perc_nonwhite 0.52  0.25
3 i_perc_white    0.48  0.25
4 i_perc_black    0.17  0.2
5 i_perc_hisp     0.18  0.18
6 i_perc_asian    0.04  0.06

```

```

# -----
# Level 2: CBSA-level variables
# -----

# Keep one observation per CBSA
cbsa_data <- data_mlca |>
  distinct(cbsa_id, .keep_all = TRUE)

# Categorical Level-2 variable
l2_categorical_desc <- cbsa_data |>
  count(cbsatype, name = "N") |>
  mutate(Proportion = N / sum(N)) |>
  select(cbsatype, Proportion) |>
  mutate(Proportion = round(Proportion, 2))

l2_categorical_desc

```

```

  cbsatype Proportion
1         0         0.48
2         1         0.52

```

```

# Continuous Level-2 variables
l2_cont <- c("inst_num", "tot_pop", "pop_density", "perc_ba",
            "med_income", "perc_pov", "perc_unemployed",
            "perc_white", "perc_black", "perc_hisp", "perc_asian")

```

```

l2_continuous_desc <- cbsa_data |>
  summarise(across(all_of(l2_cont),
    list(Mean = ~mean(., na.rm = TRUE),
         SD   = ~sd(., na.rm = TRUE)))) |>
  pivot_longer(everything(),
    names_to = c("Variable", ".value"),
    names_pattern = "(.+)_ (Mean|SD)$") |>
  mutate(across(c(Mean, SD), ~round(., 2)))

l2_continuous_desc

```

```

# A tibble: 11 x 3
  Variable      Mean      SD
  <chr>        <dbl>   <dbl>
1 inst_num      4.49    11.8
2 tot_pop       4.24    12.6
3 pop_density  191.    280.
4 perc_ba       28.6     9.29
5 med_income  69572.  14640.
6 perc_pov      9.82     3.81
7 perc_unemployed 2.92     0.84
8 perc_white    70.0    18.6
9 perc_black     8.96    11.8
10 perc_hisp    13.0    15.1
11 perc_asian     2.29     3.53

```

```

# -----
# Sample sizes
# -----

sample_sizes <- tibble(
  Level = c("Institutions", "CBSAs"),
  N = c(nrow(data_mlca), nrow(cbsa_data)))

sample_sizes

```

```

# A tibble: 2 x 2
  Level      N
  <chr>    <int>

```

```
1 Institutions 3133
2 CBSAs       725
```

```
# -----
# Save descriptive statistics
# -----

write_xlsx(list("L1 categorical" = l1_categorical_desc,
               "L1 continuous"   = l1_continuous_desc,
               "L2 categorical"  = l2_categorical_desc,
               "L2 continuous"   = l2_continuous_desc,
               "Sample sizes"    = sample_sizes),
           "../data/MLCA_ds.xlsx")
```

```
# Level-1 indicators (Y)
## cf. these are not predictors, so technically we don't need to check for multicollinearity
## cf. these are categorical variables, so VIF may not be appropriate
y_data <- data_mlca |> select(all_of(Y))

set.seed(1234)
y_data$dummy_y <- rnorm(nrow(y_data))

vif_y_model <- lm(dummy_y ~ ., data = y_data)
vif(vif_y_model)
```

public	size	awardlev	focus	research	online
1.760092	2.488298	2.215250	1.349799	2.272521	1.397997
evening	selectivity	pell	msi	nonwhite	
1.132332	1.959964	1.438918	1.491228	1.616924	

```
# Level-1 covariates (Z)
z_data <- data_mlca |>
  select(all_of(Z))

set.seed(1234)
z_data$dummy_y <- rnorm(nrow(z_data))

vif_z_model <- lm(dummy_y ~ ., data = z_data)
vif(vif_z_model)
```

GVIF Df GVIF^{1/(2*Df)}

i_perc_white	5.611458	1	2.368852
i_perc_black	4.036869	1	2.009196
i_perc_hisp	3.518189	1	1.875684
i_perc_asian	1.679566	1	1.295981
housing	1.352845	1	1.163118
rurality	1.138688	2	1.033002
religious	1.214927	1	1.102237

```
# Level-2 covariates (Zh)
zh_data <- data_mlca |>
  select(all_of(Zh)) |>
  mutate(cbsatype = as.factor(cbsatype)) # switch nominal variable to factor

zh_data$dummy_y <- rnorm(nrow(zh_data))

vif_zh_model <- lm(dummy_y ~ ., data = zh_data)
vif(vif_zh_model)
```

cbsatype	inst_num	tot_pop	pop_density	perc_ba
1.489369	55.185840	51.956170	14.507700	3.756909
med_income	perc_pov	perc_unemployed	perc_white	perc_black
7.506924	4.332139	2.383692	31.438482	9.009204
perc_hisp	perc_asian			
19.233788	8.332143			

```
set.seed(1234)

model_selection <- multiLCA(
  data = data_mlca,
  Y = Y,
  iT = 6:15,
  Z = Z,
  iM = 2:8,
  id_high = c("cbsa_id"),
  Zh = Zh,
  incomplete = FALSE,
  extout = TRUE,
  verbose = TRUE,
  sequential = TRUE,
  reord = FALSE
)
```

```

Fitting single-level measurement model (step 1)...
Fitting single-level measurement model (step 1)...
Fitting single-level measurement model (step 1)...
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Fitting multilevel measurement model (step 3)...
Fitting multilevel measurement model (step 3)...

```

\$step1

	BIClow	BIChigh	AIC	ICL_BIClow	ICL_BIChigh
iT=6	62283.9	61978.02	61019.51	-	-
iT=7	62168.61	61811.5	60692.47	-	-
iT=8	61791.57	61383.23	60103.69	-	-
iT=9	61450.78	60991.22	59551.16	-	-
iT=10	61378.01	60867.22	59266.65	-	-
iT=11	61477.86	60915.85	59154.76	-	-
iT=12	61356.19	60742.95	58821.34	-	-
iT=13	61244.3	60579.83	58497.71	-	-
iT=14	61325.93	60610.24	58367.6	-	-
iT=15	61436.13	60669.22	58266.07	-	-

\$step2

	BIClow	BIChigh	AIC	ICL_BIClow	ICL_BIChigh
iT*,iM=2	60867.33	60183.84	58042.1	61910.42	60417.86

```

iT*,iM=3 60979.2 60276.68 58075.32 62316.66 60970.78
iT*,iM=4 60751.79 60030.25 57769.27 62135.21 61024.22
iT*,iM=5 60803.87 60063.3 57742.7 62169.53 61249.8
iT*,iM=6 60713.77 59954.18 57573.95 62056.67 61233.75
iT*,iM=7 61027.61 60248.98 57809.14 62274.47 61597.22
iT*,iM=8 60930.15 60132.5 57633.04 62272.32 61274.15

```

```
$step3
```

	BIClow	BIChigh	AIC	ICL_BIClow	ICL_BIChigh
iT=6,iM*	62444.77	62094.97	60998.88	63138.45	63550.71
iT=7,iM*	61352.09	60943.76	59664.22	62103.78	61952.13
iT=8,iM*	60958.47	60491.59	59028.6	61770.99	61548.56
iT=9,iM*	60836.74	60311.32	58664.88	61645.79	61575.61
iT=10,iM*	60761.1	60177.14	58347.25	61716.34	61314.49
iT=11,iM*	60698.88	60056.37	58043.04	61721.76	61138.39
iT=12,iM*	60682.6	59981.54	57784.77	61820.49	60968.09
iT=13,iM*	60920.58	60160.99	57780.77	62278.63	61205.84
iT=14,iM*	61034.27	60216.13	57652.46	62335.95	61333.29
iT=15,iM*	61041.56	60164.88	57417.76	62569.08	61132.8

```
$optimal
```

```
iT= 12
```

```
iM= 6
```

```
analytic_model <- model_selection
```

```
# analytic_model
```

```
“r optimal 12/6 #|eval: false
```

Note: This model does not run successfully

**The model above identified the optimal 12-class / 6-cluster solution
“without” covariates**

(<https://doi.org/10.1080/00273171.2025.2473935>)

**This is the actual 12/6 model with Level-1 and Level-2 covariates,
but it currently returns a “solve(): solution not found” error**

```
set.seed(1234)
```

```
analytic_model <- multiLCA( data = data_mlca, Y = Y, iT = 12, Z = Z, iM = 6, id_high =  
c("cbsa_id"), Zh = Zh, incomplete = FALSE, extout = TRUE, verbose = TRUE, sequential  
= TRUE, reord = FALSE )
```

```
analytic_model
```

```
::: {.cell}
```

```
```.r .cell-code}
```

```
mPi: lower-level class proportions/probabilities ("How prevalent is each L1 class?")
```

```
mPhi: item-response probabilities conditional on L1 class ("What does each institution clas
```

```
mPMsumX: posterior probabilities of L1 class membership, marginalized over L2 classes ("Wh
```

```
mPW: posterior probabilities of L2 class membership ("Which L2 cluster is each CBSA most l
```

```

```

```
Level 1 post-estimation
```

```

```

```
lower-level class proportions given the higher-level latent classes and the covariates
```

```
l1_proportions <- as.data.frame(analytic_model$mPi)
```

```
item-response probabilities
```

```
l1_probabilities <- as.data.frame(analytic_model$mPhi)
```

```

posterior probabilities of Level-1 class membership
l1_posterior <- as.data.frame(analytic_model$mPMsumX)

sanity check: number of posterior rows should equal number of institutions
stopifnot(nrow(l1_posterior) == nrow(data_mlca))

predicted Level-1 class = class with highest posterior probability
l1_classes <- data_mlca |>
 select(unitid, instnm, cbsa, cbsa_id) |>
 mutate(L1_Predicted_Class = max.col(as.matrix(l1_posterior),ties.method = "first"),
 L1_Max_Probability = apply(l1_posterior, 1, max))

l1_classes

check institutions by predicted level-1 class
l1_institutions_by_class <- l1_classes |>
 arrange(L1_Predicted_Class, instnm)

l1_institutions_by_class

count number of institutions in each predicted level-1 class
l1_class_counts <- l1_classes |>
 count(L1_Predicted_Class, name = "N_Institutions")

l1_class_counts

 L1_Predicted_Class N_Institutions
1 2 1
2 44 3132

:::

vU_modal: vector of posterior modal class assignments (the class with the highest posterior
vU_modal is available for some model specifications, but is not provided as the relevant L
For the multilevel model, modal assignments can be obtained from:
mPMsumX -> predicted L1 class for each institution
mPW -> predicted L2 cluster for each CBSA

```

```

Level 2 predicted clusters

posterior probabilities of Level-2 cluster membership
l2_posterior <- as.data.frame(analytic_model$mPW)

sanity check: number of posterior rows should equal number of CBSAs
stopifnot(nrow(l2_posterior) == n_distinct(data_mlca$cbsa_id))

sanity check2: number of posterior rows should equal number of CBSA IDs
stopifnot(setequal(as.character(row.names(analytic_model$mPW)), unique(as.character(data_mlca$cbsa_id))))

assign each CBSA to the cluster with the highest posterior probability
l2_clusters <- data.frame(
 cbsa_id = row.names(analytic_model$mPW),
 L2_Predicted_Class = max.col(as.matrix(l2_posterior), ties.method = "first"),
 L2_Max_Probability = apply(l2_posterior, 1, max),
)

posterior probability for each Level-2 cluster
Cluster1 = l2_posterior[, 1],
Cluster2 = l2_posterior[, 2],
Cluster3 = l2_posterior[, 3],
Cluster4 = l2_posterior[, 4],
Cluster5 = l2_posterior[, 5],
Cluster6 = l2_posterior[, 6]

CBSA-level data: one row per CBSA
cbsa_level_data <- data_mlca |>
 mutate(cbsa_id = as.character(cbsa_id)) |>
 distinct(cbsa_id, .keep_all = TRUE) |>
 select(cbsa_id, cbsa, inst_num, all_of(setdiff(Zh, "inst_num")))

attach CBSA characteristics to predicted Level-2 clusters
l2_clusters <- l2_clusters |>
 left_join(cbsa_level_data, by = "cbsa_id")

l2_clusters

```

```
count number of CBSAs and institutions in each predicted Level-2 cluster
l2_cluster_counts <- l2_clusters |>
 group_by(L2_Predicted_Class) |>
 summarise(N_CBSAs = n(), N_Institutions = sum(inst_num), .groups = "drop")

l2_cluster_counts

write_csv(l2_cluster_counts, "../data_paper2/l2_cluster_counts.csv")
```